Journal of Flow Chemistry (2025) 15:197–206
https://doi.org/10.1007/s41981-025-00358-z

FULL PAPER

Optimal conditions for Pt-catalyzed microfluidic synthesis of iodoolefins

A. N. Bulgakov[1] · A. S. Gogil'chin[2] · Mahmoud E. A. Eid[1,3] · A. A. Tereshchenko[1] · N. V. Egil[1] · T. V. Krasnyakova[1,2] · I. O. Krasniakova[1] · A. V. Soldatov[1] · A. A. Guda[1] · S. A. Mitchenko[1,2]

Received: 9 April 2025 / Accepted: 18 June 2025 / Published online: 30 June 2025
© Akadémiai Kiadó Zrt 2025

Abstract

The employment of acetylene-derived alkenyl iodides facilitates the integration of olefinic moieties into intricate molecular structures. Their synthesis is often related to catalytic gas-liquid process where traditional batch methods suffer from inefficient mass transfer upon scaling, thus limiting yields of iodoolefins production. In this study, we develop microfluidic synthesis of vinyl iodide and (*E*,*E*)-1,4-diiodo-1,3-butadiene from acetylene using Pt^{IV} iodo complexes as a catalyst in aqueous solutions. Segmented Taylor flow regime was applied to increase interfacial surface area and subsequent gas-liquid separation enables online mass-spectroscopic conversion monitoring. We varied reaction parameters to study temperature dependence of the conversion as well as influence of gas and liquid flow rates. The optimal conditions were derived from Bayesian approach and promoted better mass transfer and, thereby, higher acetylene conversion. The methodology can be readily extended to the synthesis of other small organic iodides.

Highlights

- Segmented flow regime for the PtI_4-catalyzed synthesis of iodoolefins from acetylene.
- The Bayesian approach was applied in optimizing reaction parameters.
- Optimal conditions favored synthesis of vinyl iodide and diiodobutadiene.

Keywords Microfluidic synthesis · Taylor flow · Homogeneous catalysis · Pt catalyst · Vinyl iodide · *(E,E)*-1,4-diiodobuta-1,3-diene · Bayesian optimization

Introduction

Transition metal-catalyzed cross-coupling reactions have emerged as prominent and widely used synthetic tools in the last decades [1]. A great variety of bioactive compounds, natural compounds, pharmaceuticals, optoelectronic materials has been efficiently prepared using this methodology [2, 3]. A plethora of pivotal products (naproxen, eletriptan, singulair, prosulfuron) has been successfully commercialized, while numerous others are still under development [4, 5]. Nowadays, cross-coupling reactions play an essential role in fine organic synthesis as they allow complex structures to be easily assembled from simpler building blocks [6].

Olefin moieties, especially conjugated dienes are widely employed as building blocks in organic synthesis via cross-coupling reactions [7]. Among such building blocks, vinyl iodides and (*E*,*E*)-1,4-diiodo-1,3-butadiene are the most promising for further involvement in C-C coupling reactions due to the high reactivity of C-I bonds, enabling these transformations under mild conditions, even at room temperature [8]. Moreover, cross-coupling reactions involving C-I derivatives offer a set of extra benefits such as achieving high stability of functional groups, retention of molecular configuration and asymmetric centers.

✉ A. A. Guda
guda@sfedu.ru

S. A. Mitchenko
samit_rpt@mail.ru

1 The Smart Materials Research Institute, Southern Federal University, Rostov-on-Don 344090, Russia

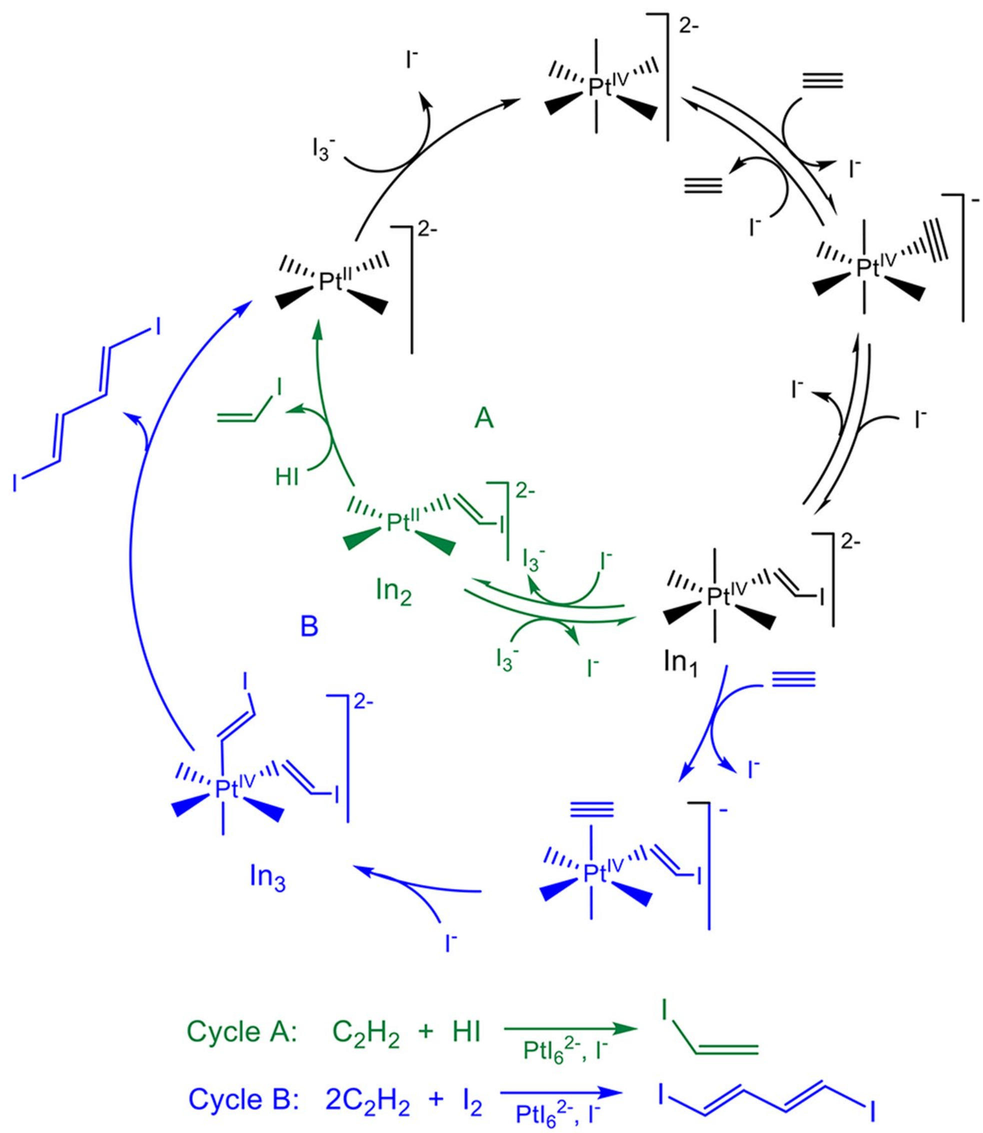
2 Institute of Physical Organic Chemistry and Coal Chemistry, 70 R. Luxemburg Str, Donetsk, Russia

3 Physics Department, Faculty of Science, Tanta University, Tanta 31527, Egypt

The synthesis of substituted diiodienes is a well-developed area, but the preparation of unsubstituted derivatives is rather challenging. The high reactivity of these substrates leads to numerous side reactions, including free-radical ones, photochemical transformations, oligomerization, deiodination [9]. Such reactions significantly reduce the yield of target products due to the rapid decomposition of diiodienes during separation and purification [7].

In earlier studies we showed how vinyl iodide [10] and (*E*, *E*)-1,4-diiodo-1,3-butadiene [11] can be synthesized with high chemo- and stereoselectivity directly from acetylene in reactions catalyzed by Pt^{IV} iodo complexes in aqueous solutions. Scheme 1 shows the mechanistic pathway of the reactions, where both involve the same monovinyl Pt^{IV} complex (In_1, Scheme 1) as a common intermediate [12, 13]. In acidic solutions, the latter is reversibly reduced by iodide ions to the intermediate vinyl Pt^{II} derivative (In_2), which protonolysis provides vinyl iodide and Pt^{II} (cycle A, Scheme 1). In neutral solutions (but in the presence of iodine in excess), further iodoplatination of the second acetylene molecule occurs by the intermediate In_1 to form the divinyl Pt^{IV} derivative (In_3). Finally, subsequent conversion of the latter via a reductive elimination step yields the C-C coupling product, (*E*, *E*)-1,4-diiodobuta-1,3-diene (cycle B, Scheme 1).

Subsequent optimization of the (*E*,*E*)-1,4-diiodo-1,3-butadiene synthesis focused on systematic variation of solvent systems, reaction temperatures, and catalyst precursors [14]. The optimal conditions were identified as room temperature reaction in acetone, which provides dual advantages: (i) excellent acetylene solubility enabling rapid substrate conversion, and (ii) pronounced solvent polarity ensuring high stereoselectivity. However, in this approach, the current workup protocol requires product isolation through aqueous precipitation at the reaction endpoint. This methodology presents two substantial drawbacks: first, it prevents reuse



Scheme 1 The stepwise mechanisms for acetylene hydroiodination (cycle A) [10] and acetylene dimerization accompanied by addition of iodine (cycle B) [11] catalyzed by Pt^{IV} iodo complexes in aqueous solutions

of the reaction medium for subsequent batches, and second, it necessitates an elaborate catalyst regeneration procedure as only the catalytic component can be effectively recovered. These limitations significantly impact the process efficiency and scalability.

A significant limitation inherent to batch reactor systems involves the precise determination of interfacial area under conditions of vigorous stirring or bubbling, where rapid diffusion processes occur [15]. To obtain a well-defined specific area for estimation of mass-transfer and reaction kinetics dependency on it, a Lewis's type stirred reactor with a plain interface between gas and liquid volumes can be used. However, the slow diffusion of gas into solution restricts the efficiency of the method.

Microfluidic synthesis, characterized by fluid flow in tiny channels with characteristic dimensions ranging from tens to hundreds of micrometers, has emerged as a transformative approach in modern chemical engineering [16]. This approach involves manipulating small volumes of fluids in microchannels and provides several benefits [17, 18] including improved mass transfer [19], online monitoring of reaction conditions [20], low thermal inertia and the ability to perform high-throughput screening of reaction conditions [21]. In addition, microfluidic devices are generally easy to use and manufacture, as well as cost effective [22], making them particularly attractive for both laboratory-scale research and potential industrial applications.

The implementation of microfluidic continuous-flow methodology for iodoolefin synthesis would overcome current limitations through two principal mechanisms: first, by intensifying acetylene mass transfer to the catalytic phase via enhanced interfacial contact [23], and second, by enabling precise quantification of specific interfacial area within the capillary reactor [24].

In this work, we present a comprehensive strategy for optimizing product yields and reducing reaction times in the homogeneous catalytic synthesis of iodoolefins using Pt^{IV} iodo complexes. We did not investigate the possibility of reusing the reaction medium or the need for catalyst regeneration, these challenges remain open. Our approach integrates four key components: (a) translation of batch reactions to segmented gas-liquid flow conditions, (b) real-time droplet monitoring to track interfacial area, (c) inline phase separation for efficient product isolation, and (d) direct monitoring of acetylene conversion kinetics via in situ mass spectrometry (MS).

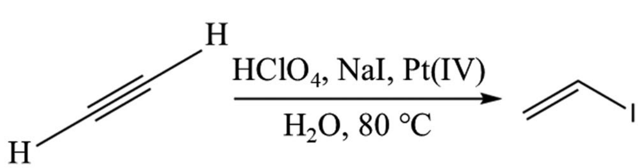


Fig. 1 Reaction scheme of vinyl iodide synthesis from acetylene

Experimental section

Reagents *Potassium hexachloroplatinate (K_2PtCl_6), perchloric acid ($HClO_4$), sodium iodide dihydrate ($NaI{\cdot}2H_2O$), crystalline iodine. All the reagents above were purchased from Sigma-Aldrich and used without further purification. Also, distilled water (Milli-Q, Millipore, 18.2 MΩ·cm), argon (99.999% purity) and 10% C_2H_2/Ar mixture were used.*

Reaction mixture *Vinyl iodide was synthesized from acetylene according to the scheme shown in Fig. 1. First, 4.05 g (21.78 mmol) of $NaI{\cdot}2H_2O$ was dissolved in distilled water to a volume of 7.6 mL followed by the addition of 2.6 mL (30.14 mmol) of 70% $HClO_4$. The obtained solution was purged with pure argon for 5 min to remove air oxygen from the reaction mixture. Then, 0.04 g of K_2PtCl_6 (823 mmol) was added and the solution was stirred for 40 min at 400 rpm using magnetic stirrer. The transparent solution turned into a dark red color due to the replacement of platinum chloride ligands with iodide ligands (Eq. 1).*

$$PtCl_6^{2-} \xrightarrow{I^-} PtI_6^{2-} \tag{1}$$

Similar reaction mixtures were prepared for both the experiment in batch and flow conditions.

To prepare reaction mixture for synthesis of (*E*, *E*)-1,4-diiodobuta-1,3-diene, 2 g (10.75 mmol) of $NaI{\cdot}2H_2O$, and 1.8 g of crystalline iodine were dissolved in distilled water to give a 10 mL solution. The solution was stirred for 60 min at 400 rpm on a magnetic stirrer. Then, 0.034 g (699.5 mmol) of K_2PtCl_6 was added.

Each experiment was performed as follows: firstly, the entire system was purged with a 10% C_2H_2/Ar gas mixture and pressurized to 3 bar. The capillary reactor, immersed in an oil bath, was then heated to the target temperature (e.g., 80 °C). At this stage, a reference mass spectrometry signal (I_g) was acquired for the pure acetylene/argon mixture to establish the baseline for zero conversion. Subsequently, the liquid catalyst was introduced by a syringe pump (Smart Microfluidics, Southern Federal University) to establish a Taylor flow regime [25], characterized by alternating gas and liquid segments. The gas flow was controlled via EL-FLOW mass-flow controllers (Bronkhorst High-Tech B.V, Ruurlo, the Netherlands). The gas phase after separation was continuously analyzed by mass spectrometry (Pfeiffer Vacuum + Fab Solutions, Aßlar, Germany) (I_{g+l}). Acetylene

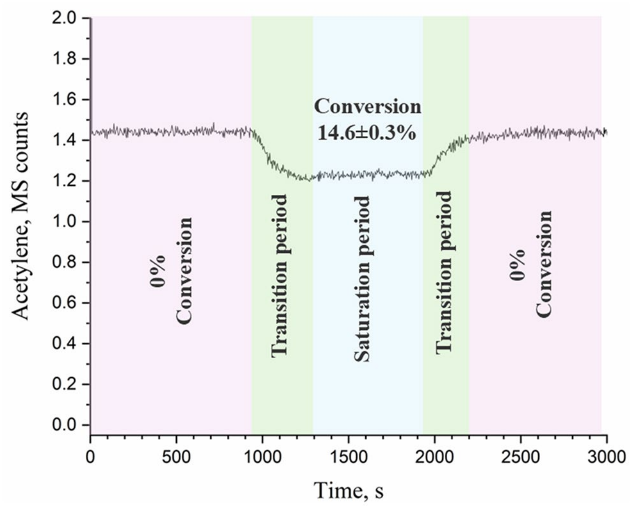


Fig. 2 In situ measured acetylene signal at the outlet of microfluidic reactor using MS

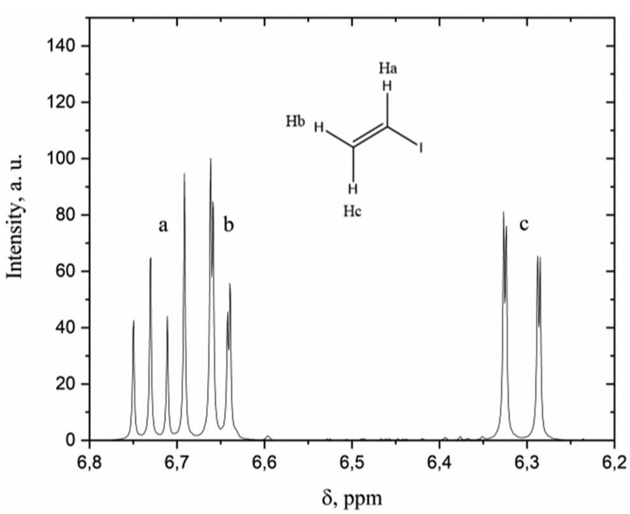


Fig. 3 Vinyl iodide, [1]H NMR (in acetone-d$_6$, ppm): H$_a$ 6.72 (dd, J^3_{H-H} (trans) = 15.5, 7.7 Hz, 1H); H$_b$ 6.65 (dd, J^3_{H-H}(cis) = 7.7, J^2_{H-H} (gem) 1.2 Hz, 1H); H$_c$ 6.31 (dd, J^3_{H-H} (trans) 15.5, J^2_{H-H} (gem) 1.2 Hz, 1H)

conversion was determined by monitoring the decay of its characteristic MS signal intensity (Eq. 2):

$$\text{Conversion } (C_2H_2) = (1 - \frac{I_{g+l}}{I_g}) * 100\% \qquad (2)$$

Then, after 10–15 min of signal accumulation, the liquid flow was stopped, and a system was flushed with a 10% C_2H_2/Ar gas mixture. Such an operation was repeated 3 times for each measurement. A plan of experiment is shown in Fig. S2, ESI†.

Batch synthesis *Instead of capillary reactor* ***(5)*** *shown in Fig. 5, a test tube filled with reaction mixture was used (Fig S3, ESI†). Reaction mixture was purged with 10% C_2H_2/Ar mixture for 20 min with gas flow rate in the range 1.6–3.2 mL/min via the tube immersed right down to the bottom of the test tube (total volume of test tube was 2.5 mL; height of liquid column was ca. 5 cm).*

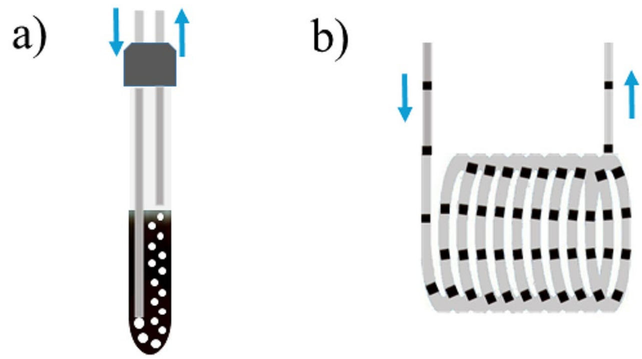


Fig. 4 Bubble column (**a**) versus segmented-flow (**b**) types of gas liquid reaction

The standard procedure of MF synthesis consists of calibration, transition mode and saturation of acetylene signal (Fig. 2). Since the dead volume of the microfluidic setup is much smaller, the C_2H_2 mass spectral signal reached saturation in ca. 5 min, much faster than in the batch experiment. All data obtained with variation of gas and liquid flows are summarized in Table S1, ESI†.

The gaseous products of the reaction can be also monitored by FTIR spectroscopy (see details in Sect. 2, ESI†), however this approach involves larger volume of the transmission cell and thus was not applied for in situ measurements.

Results and discussion

Comparison between batch and flow syntheses

To verify the performance of the catalyst, the first experiment was carried out under batch conditions at gas flow rate of 2 mL/min during 90 min (Fig. S4, ESI). The total conversion equal 9.4±0.5% C_2H_2 was reached. We collected in total 3 mg of vinyl iodide during the experiment and verified the purity by ^{1}H and ^{13}C{^{1}H} NMR in acetone-D$_6$ solution using benzene as an internal weight standard.

Conventional flask-based synthesis (see Fig. 4a), though common, faces some limitations. Gas bubbles in the bulk volume have different sizes and nonuniform concentration in the liquid. At higher gas flow rate the bubbles tend to coalescence, thus lowering reaction rates and making difficult quantitative analysis of reaction kinetics [26].

Segmented flow regime (see Fig. 4b) overcome bubble column limitations by enabling precise gas-liquid interface. Segmented gas-liquid flow improves solubility [27], enhances mass transfer, and thus increase conversion rates [28]. This work aims to optimize the conditions of gas-liquid catalytic reaction within capillary microreactor.

Microfluidic synthesis *A scheme of the flow setup is shown in Fig. 5. The reaction mixture (5.5 mL) was loaded into glass syringes installed into syringe pumps* ***(1)*** *(see also Fig. S1, ESI†). Then, reaction mixture (flow rates 0.06–0.18 mL/min) was injected into the flow of 10%* C_2H_2*/Ar (1.6–3.2 mL/min), dosed by mass flow controller* ***(2, 3)****, using PEEK T-junction for gas-liquid segmented flow generation* ***(4)****. Reaction time varied from 5 to 10 min depending on the flow rates. Generated liquid segments were sent into capillary reactor* ***(5)*** *— a FEP tube with an inner diameter of 0.8 mm, outer diameter of 1.6 mm, and length of 5 m (inner volume ca. 2.5 mL) kept at 80 °C using oil bath and heater. After the capillary, the velocity of the liquid segments was recorded on a microscope with camera* ***(6)****. Then, gas and liquid phases were separated using PTFE Y-connector* ***(7)*** *to avoid contact of mass spectrometer with liquid phase. The whole system was pressurized with 3 bars maintained using a low-flow back-pressure regulator* ***(9)*** *and pilot pressure from cylinder* ***(10)****. The liquid phase was collected using another syringe pump* ***(11)*** *to ensure that only the gas phase is analyzed by the mass spectrometer* ***(12)****. The products in the gas phase were condensed and collected in the test tube cooled with liquid nitrogen* ***(13)****. The microfluidic approach has significant potential for scaling [29, 30], but its implementation depends on the specific system architecture and the tasks at hand. Microfluidic gas lines can be integrated into multichannel arrays, enabling increased synthesis throughput. Microfluidic technologies scale well provided the setup is optimized and key synthesis parameters (pressure, temperature) are maintained. Parallel microreactors with slightly larger tube diameters can serve as a good scaling model for gas-liquid systems. A photo of the whole experimental setup is shown on Fig. S1, ESI†.*

Screening of reaction parameters in microfluidic regime

To systematically evaluate the influence of key reaction parameters, we conducted a series of experiments varying reaction temperature and liquid-phase flow rate while maintaining constant gas flow conditions. The reaction temperature was varied from 60 °C to 100 °C at a fixed pressure of 3 bar and constant gas and fluid flow rates (2 and 0.045 mL/min, respectively). The acetylene conversion grew monotonically as the temperature increased (Fig. 6). The temperature of 80 °C was selected as the optimal setting for further optimization of the other reaction parameters. At this temperature, acetylene conversion is high enough for detection, while the selectivity for vinyl iodide remains almost 100% [10]. The apparent activation energy of acetylene hydroiodination reaction obtained from the temperature dependence of acetylene conversion (Fig. 12) was found to be of 27.3 ± 1.9 kJ*mol^{-1}.

The gas-liquid ratio represents another optimization parameter in microfluidic synthesis, exerting significant influence on reaction efficiency. As demonstrated by Dengwei et al. [31], the Taylor flow enhances mass transfer

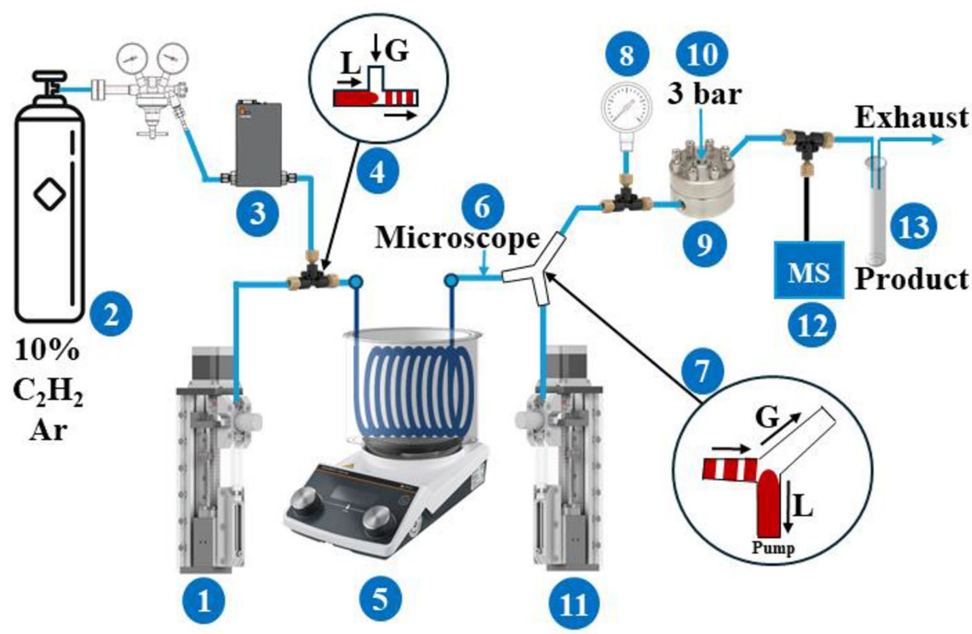


Fig. 5 A scheme of experimental setup used for synthesis in microfluidic conditions L– liquid flow, G– gas flow.

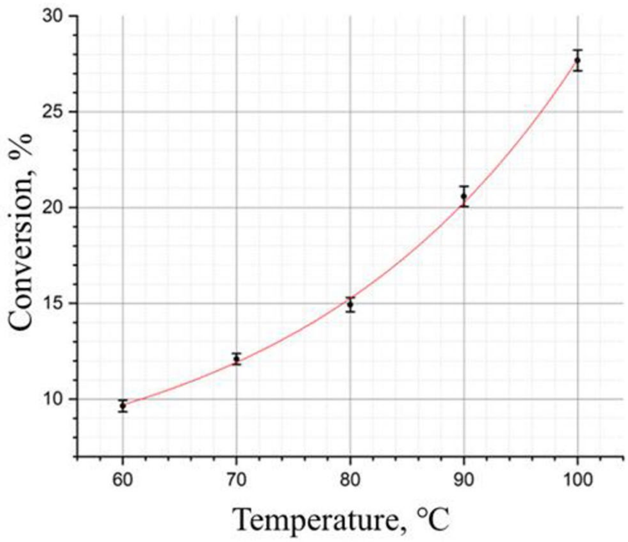


Fig. 6 Temperature dependence of the conversion during microfluidic synthesis of vinyl iodide in segmented flow regime

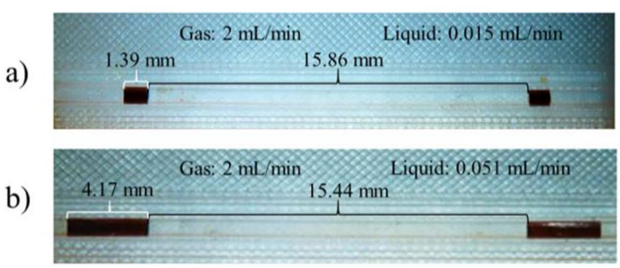


Fig. 7 Resulted patterns of liquid segments with liquid flow variation. The MF system was pressurized under 3 bar. Reaction temperature was 80 °C

through increased direct gas-liquid contact area between segments and via film that forms from the liquid on the capillary wall. We conducted a series of experiments where gas and liquid flow rates were varied independently (see Figs. S8–10, ESI†). Higher liquid flow rate led to elongation of the liquid segments (see Fig. 7) while lower liquid flow rate at a fixed gas flow rate (see Fig. S8) resulted in longer gas segments and narrower liquid segments. Fixed width of liquid segments can be achieved by simultaneous adjustment of both gas and liquid flow rates (see Fig. S9) [32]. Finally, reducing the gas flow rate at constant liquid flow rate (see Fig. S10) results in wider liquid segments and shorter gas segments.

The relationship between observable length of liquid and gas segments with conversion is shown in Fig. 8a. The optimal parameters can be determined by the size of orange circles that are proportional to the conversion. Amount of vinyl iodide in the gas phase increases along with the length of liquid segments and for shorter gas segments. The highest achieved conversion corresponds to the flow pattern with 3 mm liquid segments and 10 mm gas bubbles. This result can be explained by two factors, both mass transfer and reaction time that depends mostly on the gas flow rate.

To explain conversion dependence on the segment sizes we calculated the overall mass transfer coefficient (Eq. 3), using approach described by Tan et al. [19].

$$K_L = K_{L_0} * \frac{\rho\, du^2}{\gamma} * \left(\frac{Q_L}{(Q_L + Q_G)}\right)^{-0.3} \tag{3}$$

where $K_{L0} = 4.03 \times 10^{-1}$ m/s; ρ is the density of water phase, 1356 kg/m3 at 20 ˚C; d is the hydrodynamic diameter, m; u is the velocity of the superficial velocity of the liquid

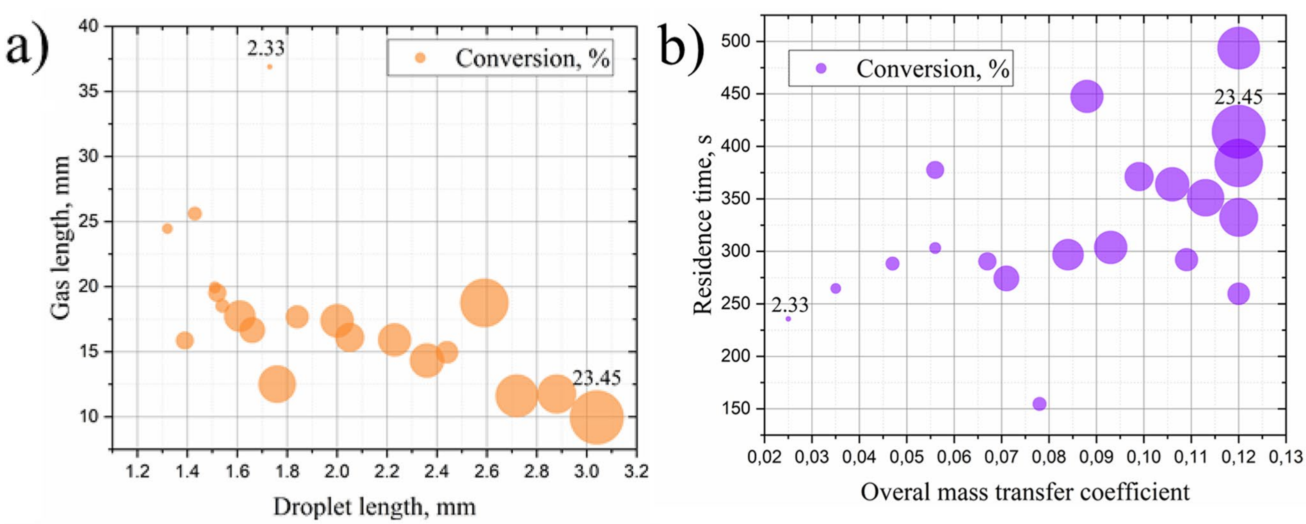


Fig. 8 Conversion variation shown as a function of gas/liquid segment sizes (**a**) or time and mass transfer coefficient (**b**). The area of circles is proportional to the conversion. All experiments were performed at T = 80 °C

calculated by liquid flow rate Q_L, m/s; gas flow rate Q_G, m/s; γ is interfacial tension between gas and liquid phase.

Figure 8b shows the conversion as a function of mass transfer and residence time. The point with lowest conversion 2.33% in panel a) is characterized by smallest mass transfer coefficient. On contrary experiments with highest conversion can be characterized with optimal mass transfer values. Results in Fig. 8 indicate that shorter gas segments provided highest efficiency. This can be explained both by the increased residence time and better replacement of the fresh gas atmosphere near liquid interface.

Residence time, representing the mean duration of reactant exposure within the reactor, serves as a parameter governing conversion efficiency. It is influenced by both the liquid and gas flow rates. As evidenced by the flow rate dependence studies (see Sect. 4, ESI †) conversion efficiency decreases systematically with increasing liquid flow rate due to reduced contact time. The optimal flow rates are those that achieve a balance between maximizing conversion and minimizing the residence time, representing an efficient and productive process. In this case the optimal value of overall mass transfer coefficient is 0.12. The optimal residence time was found to be about 7 min, which ensured high acetylene conversion for not too long exposure time.

Optimization of liquid and gas flow rates

Based on preliminary observations in the previous section we systematically optimize the liquid and gas flow rates, using a Bayesian approach. We followed a stepwise procedure. First the liquid flow rate was varied while maintaining a constant gas flow rate (2 mL/min) (see Fig. 9a). In a second step we generated uniformly distributed sampling points across both liquid and gas flows using Improved Latin Hypercube Sampling (IHS) [33, 34]. This approach minimized blind zones in the conversion mapping, ensuring comprehensive coverage of the design space (see Fig. 9b). Finally, the optimal parameters for maximum conversion were discovered during Bayesian optimization algorithm. All experiments utilizing the Bayesian optimization algorithm are presented. The algorithm itself was employed only in the third step (see Fig. 9c) and reached its optimum at the 4th point. Additionally, we have included the source code for the Bayesian optimization in ESI (Sect. 3). The acquisition function was chosen Upper Confidence Bound (UCB, see Fig. S7, ESI†) to maximize the target function [35] (Eq. 4):

$$UCB = \mu\ (x) + k^{*}\sigma\ (x) \tag{4}$$

where μ(x) — prediction of the target function (picture in the center), k = 2.56 — coefficient influencing the optimizer's work in the modes of search or refinement, empirically selected for some balance between the modes, σ(x) — standard deviation.

The background color corresponds to the prediction of the total conversion performed by radial basis function (RBF) algorithm trained on available experimental points in each step. The coefficient of determination (R^2) was computed using the Leave-One-Out Cross-Validation method, serving as a metric for the model's goodness of fit.

The implementation of Bayesian optimization serves two primary objectives: systematic reduction of experimental iterations while maintaining convergence to optimal conditions, and efficient maximization of the target objective function. This machine learning strategy achieves these goals through intelligent balancing of exploration of uncertain parameter spaces and exploitation of known high-performance regions. Using such an approach for liquid and gas flow rates, we achieved 23.5 ± 0.8% conversion in the microfluidic regime.

Reaction with other iodoolefins

To perform a (*E*,*E*)-1,4-diiodobuta-1,3-diene synthesis (Fig. 10) in flow conditions, the same setup was used (Fig. 4). The optimal parameters obtained for synthesis of vinyl iodide were found to be suitable for related reaction and the ^{1}H and $^{13}C\{^{1}H\}$ NMR spectra confirm that (*E*, *E*)-1,4-diiodobuta-1,3-diene is successfully formed in the microfluidic system. The temperature dependence of the conversion reveals similar values of apparent activation energies for both reactions (Fig. 12). This coincidence of the two reaction barriers would seem to be the result of their limitation by the step of acetylene iodoplatination to form the In1 intermediate, a monovinyl derivative of Pt(IV). However, this is not a case: at room temperature, we observed the equilibrium formation of the only organometallic intermediate In_1 [12–13]. This is indicative of the acetylene hydroiodination reaction limitation by the reduction of the In1 intermediate to the vinyl derivative of Pt(II), In_2, followed by a fast protonolysis of the latter to release vinyl iodide (Scheme 1, Cycle A). Therefore, we believe that the observed matching of the two reaction barriers is purely coincidental.

Both synthetic processes are based on atom-economical addition reactions, and both can be carried out in aqueous solutions. Both of these features are particularly important in light of the current green chemistry demands. The demonstrated approach can be extended to optimize the reactions of similar substrates (e.g., methyl acetylene, pentyne-1, etc.). In addition, various catalytic reaction parameters can be easily varied to further increase the substrate conversion.

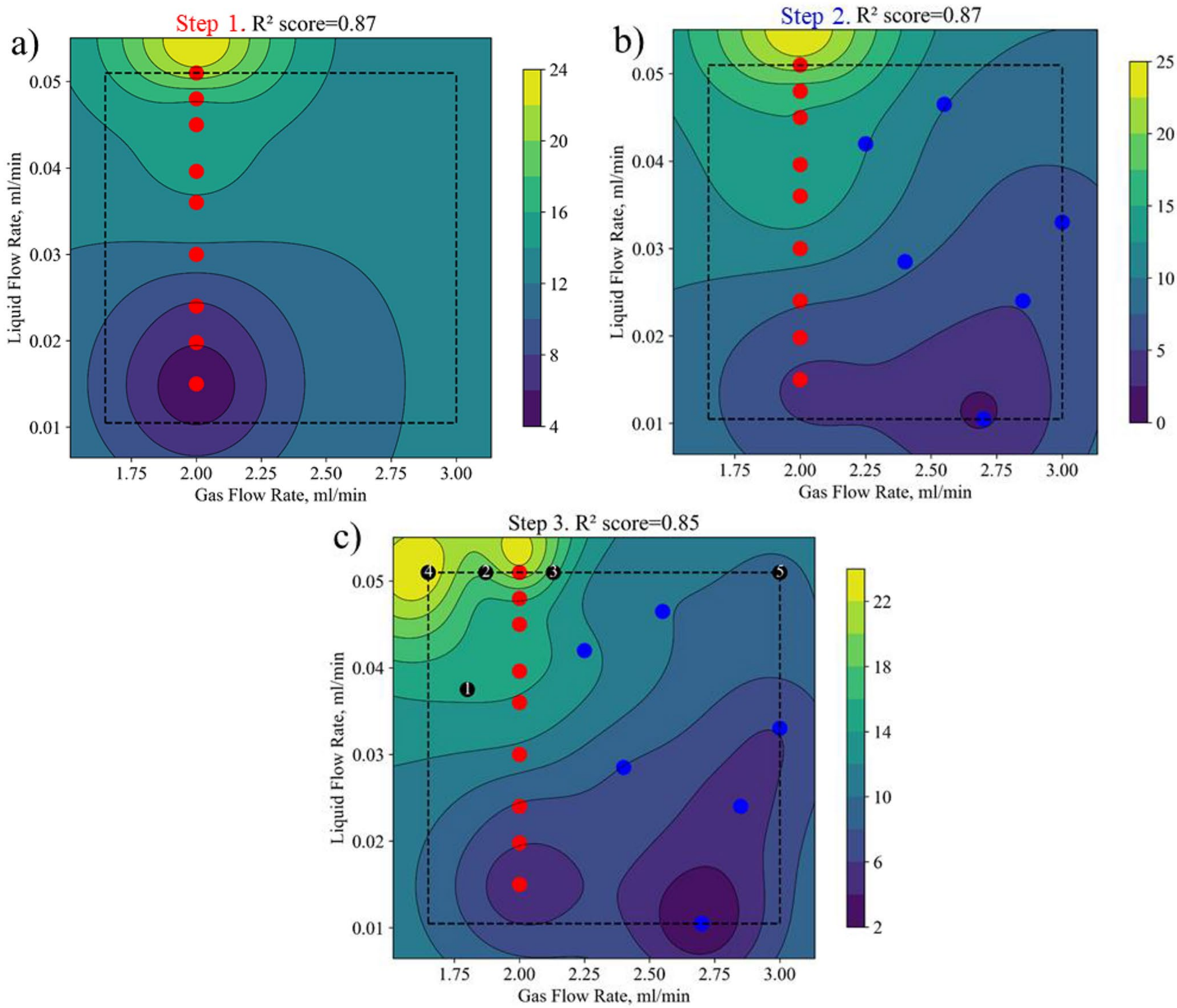


Fig. 9 Three-step optimization in the space of gas and liquid flows reaction parameters. Red points were selected in the first step; blue points correspond to IHS mapping in the second step selection. Finally, black points were selected one by one in Bayesian algorithm. The background color in each panel correspond to the prediction of conversion by RBF algorithm trained on available experimental points

Table 1 Setup productivity of MF and batch regime

	Conversion, %	Space-time-yield, mL/min	$\nu(C_2H_3I)$, mol/(min*g)
MF regime	23.5 ± 0.8	0.39 ± 0.013	$1.186*10^{-4} \pm 4.08*10^{-6}$
Batch regime	9.4 ± 0.5	0.19 ± 0.01	$4.742*10^{-5} \pm 2.52*10^{-6}$

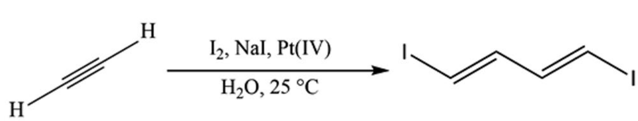


Fig. 10 Reaction scheme of (*E*, *E*)-1,4-diiodobuta-1,3-diene synthesis from acetylene

Conclusion

The synthesis of vinyl iodide and (*E*, *E*)-1,4-diiodobuta-1,3-diene has been successfully transferred into microfluidic regime. The implementation of Bayesian optimization provided efficient optimization of the reaction parameters inside coil reactor and increased overall acetylene conversion to 23.5 ± 0.8% for vinyl iodide within less than 10 min. C_2H_2 conversion increased with larger liquid segments and shorter gas segment length with the higher values achieved when the length of the gas segment was about triple the length of the liquid counterpart. The overall mass transfer coefficient at the optimal conditions was 0.12. The

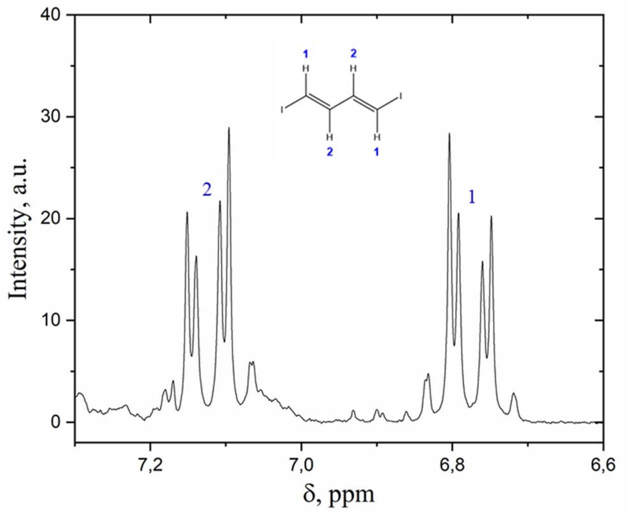


Fig. 11 *E*, *E*-1,4-diiodobuta-1,3-diene, ^{1}H NMR (acetone-d_6, δ): 7.06–7.17 (m, 2 H), 6.72–6.82 (m, 2 H). $^{13}C\{^{1}H\}$ (CD_3OD): δ = 145.5 (-CH=), 83.1 (-CI=)

Table 2 List of conducted experiments for (*E*, *E*)-1,4-diiodobuta-1,3-diene synthesis with temperature variation

#	Liquid Flow Rate, mL/min	Gas Flow Rate, mL/min	Temperature, °C	Acetylene conversion, %
1	0.03	2.2	20	3.8 ± 0.2
2	0.03	2.2	35	7.1 ± 0.2
3	0.03	2.2	50	10.5 ± 0.2

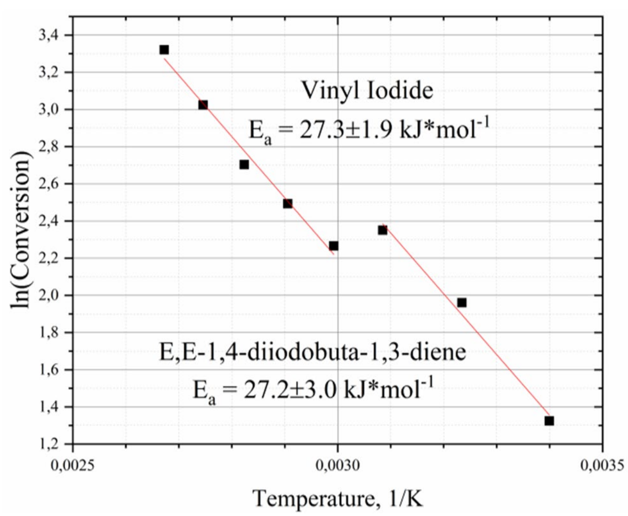


Fig. 12 Arrhenius plot for vinyl iodide and (*E*, *E*)-1,4-diiodobuta-1,3-diene

optimized flow rates allowed to achieve 10.5% conversion for the (*E*, *E*)-1,4-diiodobuta-1,3-diene already at 50 °C.

The microfluidic approach coupled with Bayesian optimization opens new perspectives for optimizing catalytic reactions and achieve rational use of reagents under variable conditions. The demonstrated approach can be extended to the optimization of many other industrially relevant reactions and catalytic systems.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s41981-025-00358-z.

Author contributions The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Funding The research was financially supported by the Strategic Academic Leadership Program of the Southern Federal University ("Priority 2030").

Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Code availability The code used in the current study are available from the corresponding author on reasonable request.

Declarations

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Consent for publication All authors have reviewed and approved the final version of this manuscript and agree to its submission and publication. This study does not include identifiable human data, images, or other personal information, and therefore, no additional consent for publication is required.

Consent to participate This study did not involve human participants, and therefore, no consent to participate was required. All experiments and data collection were conducted in accordance with ethical guidelines and institutional regulations.

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

1. Patel IS, Ganesan G, Jain S (2024) Catalytic advancements: optimizing pd-based cross-coupling reactions through flow chemistry. angewandte chemie international edition 53, 13001–13002 org. process res. dev. 28: 3464–3508. https://doi.org/10.1021/acs.oprd.4c00027
2. Fairlamb IJS (2014) Metal-Catalyzed Cross-Coupling reactions and more. Angew Chem Int Ed 53:13001–13002. https://doi.org/10.1002/anie.201408267
3. Magano J, Dunetz JR (2011) Large-Scale applications of transition Metal-Catalyzed couplings for the synthesis of pharmaceuticals. Chem Rev 111:2177–2250. https://doi.org/10.1021/cr100346g
4. Corbet J-P, Mignani G (2006) Selected patented Cross-Coupling reaction technologies. Chem Rev 106:2651–2710. https://doi.org/10.1021/cr0505268

5. Schlummer B, Scholz U, Palladium-Catalyzed C–N (2004) C–O Coupling–A practical guide from an industrial vantage point. Adv Synth Catal 346:1599–1626. https://doi.org/10.1002/adsc.200404216
6. Tabassum S, Zahoor AF, Ahmad S (2022) Cross-coupling reactions towards the synthesis of natural products. Mol Divers 26:647–689. https://doi.org/10.1007/s11030-021-10195-6
7. Ananikov VP, Hazipov OV, Beletskaya IP (2011) 1,4-Diiodo-1,3-dienes: versatile reagents in organic synthesis. Chem– Asian J 6:306–323. https://doi.org/10.1002/asia.201000405
8. Knochel P, Cross-Coupling Reactions (2003) A practical guide. Adv Synth Catal 345:1031–1031. https://doi.org/10.1002/adsc.200303073
9. Yang W, Guo J, Hee S, Chen Y (2025) Recent advances in Iodine-Mediated radical reactions. Adv Synth Catal 367:e202401486. https://doi.org/10.1002/adsc.202401486
10. Mitchenko SA, Zamashchikov VV, Shubin AA (1993) The kinetics and mechanism of reactions of acetylene in aqueous iodide solutions of platinum complexes. 2. Hydroiodation of acetylene of Pt(IV)-NaI-H^+-H_2O system. *Kinetika i Kataliz*, 34(3), 421–425., ISSN: 0023-1584
11. Mitchenko SA, Ananikov, Beletskaya VP (1997) A novel stereoselective and catalytic CC coupling reaction: acetylene dimerization accompanied by addition of iodine to yield (E,E)-1,4-diiodobuta-1,3-diene in the Pt^{IV}–I^-–I_2–MeOH system. Mendeleev Commun 7:130–131. https://doi.org/10.1070/MC1997v007n04ABEH000715
12. Mitchenko SA, Zamashchikov VV, Shubin AA (1993) The kinetics and mechanism of acetylene reactions in aqueous iodide solutions of platinum complexes.1. Reversible stages of iodoplatination of acetylene in Pt(IV)-I–H_2O solutions. Kinet Katal 34(2):222–225
13. Mitchenko SA, Zamashchikov VV, Pryadko ON Registration of PMR spectra of platinum(IV) beta-iodovinyl complex as intermediate of acetylene hydroiodation in Pt^{IV}-I–H_2O system. Zh Obshch Khim 65(2), 346–346, ISSN: 0044-460X (1995).
14. Ananikov VP, Kashin AS, Hazipov (2011) Highly selective catalytic synthesis of (E,E)-1,4-Diiodobuta-1,3-diene via Atom-Efficient addition of acetylene and iodine: A versatile (E,E)-1,3-Diene Building block in Cross-Coupling reactions. Synlett 22:2021–2024. https://doi.org/10.1055/s-0030-1261166
15. Gorbunov D, Nenasheva M, Shashkin (2024) G transferring hydroformylation reaction into high-pressure gas–liquid microfluidic systems: key achievements and perspectives. J Ind Eng Chem 136:46–72. https://doi.org/10.1016/j.jiec.2024.02.029
16. Armstrong CD, Teixeira AR (2020) Advances in dynamically controlled catalytic reaction engineering. Reaction Chem Eng 5:2185–2203. https://doi.org/10.1039/D0RE00330A
17. Tehranirokh M, Kouzani AZ, Francis PS (2013) Microfluidic devices for cell cultivation and proliferation. Biomicrofluidics 7 Online ISSN 1932– 1058. https://doi.org/10.1063/1.4826935
18. Shrimal P, Jadeja G, Patel S (2020) A review on novel methodologies for drug nanoparticle preparation: microfluidic approach. Chem Eng Res Des 153:728–756. https://doi.org/10.1016/j.cherd.2019.11.031
19. Tan J, Lu YC, Xu JH (2012) Mass transfer performance of gas–liquid segmented flow in microchannels. Chem Eng J 181–182:229–235. https://doi.org/10.1016/j.cej.2011.11.067
20. Hsieh C-H, Chao C-S, Mong K-K (2012) T. Online monitoring of chemical reactions by contactless atmospheric pressure ionization mass spectrometry. J Mass Spectrom 47:586–590. https://doi.org/10.1002/jms.2983
21. Churski K, Korczyk P, Garstecki P (2010) High-throughput automated droplet microfluidic system for screening of reaction conditions. Lab Chip 10:816–818. https://doi.org/10.1039/B925500A
22. Scott SM, Ali Z (2021) Fabrication methods for microfluidic devices: An overview. Micromachines 12(3):318. https://doi.org/10.3390/mi12030319
23. Xu H, Xu C, Zheng Y (2024) Continuous synthesis of a silver-based primary explosive with high purity. Chem Eng Sci 300:120634. https://doi.org/10.1016/j.ces.2024.120634
24. Parra-Cabrera C, Achille C, Kuhn S (2018) 3D printing in chemical engineering and catalytic technology: structured catalysts, mixers and reactors. Chem Soc Rev 47:209–230. https://doi.org/10.1039/C7CS00631D
25. Taylor GI (2006) Deposition of a viscous fluid on the wall of a tube. J Fluid Mech 10:161–165. https://doi.org/10.1017/S0022112061000159
26. Fu T, Ma Y (2015) Bubble formation and breakup dynamics in microfluidic devices: A review. Chem Eng Sci 135:343–372. https://doi.org/10.1016/j.ces.2015.02.016
27. Porta R, Benaglia M, Puglisi A, Flow Chemistry (2016) Recent developments in the synthesis of pharmaceutical products. Org Process Res Dev 20:2–25. https://doi.org/10.1021/acs.oprd.5b00325
28. Plouffe P, Macchi A, Roberge DM (2014) From batch to continuous chemical Synthesis—A toolbox approach. Org Process Res Dev 18:1286–1294. https://doi.org/10.1021/op5001918
29. Dong Z, Wen Z, Zhao F (2021) Scale-up of micro- and millireactors: an overview of strategies, design principles and applications. Chem Eng Science: X 10:100097. https://doi.org/10.1016/j.cesx.2021.100097
30. Kockmann N, Gotsponner M, Roberge DM (2011) Scale-up concept of single-channel microreactors from process development to industrial production. Chem Eng J 167:718–726
31. Dengwei F, Yifei H, Zhaoyu L (2024) An experimental investigation on characteristics of liquid film thickness of gas-liquid Taylor flow in square/rectangular microchannel applied in microreactor. Int J Heat Mass Transf 234:126081. https://doi.org/10.1016/j.ijheatmasstransfer.2024.126081
32. Günther A, Khan SA, Thalmann M (2004) Transport and reaction in microscale segmented gas–liquid flow. Lab Chip 4:278–286. https://doi.org/10.1039/B403982C
33. Shapovalov VV, Chapek SV, Tereschenko AA (2023) 3D-printed microfluidic system for the in situ diagnostics and screening of nanoparticles synthesis parameters. Micro Nano Eng 20:100224. https://doi.org/10.1016/j.mne.2023.100224
34. Egil NV, Zagrebaev AD, Medvedev PV (2024) High-throughput screening of gold nanoparticle synthesis parameters in droplet microfluidics. Mendeleev Commun 34:783–785. https://doi.org/10.1016/j.mencom.2024.10.005
35. Shahriari B, Swersky K, Wang Z (2016) Taking the human out of the loop: A review of bayesian optimization. Proc IEEE 104:148–175. https://doi.org/10.1109/JPROC.2015.2494218

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.